

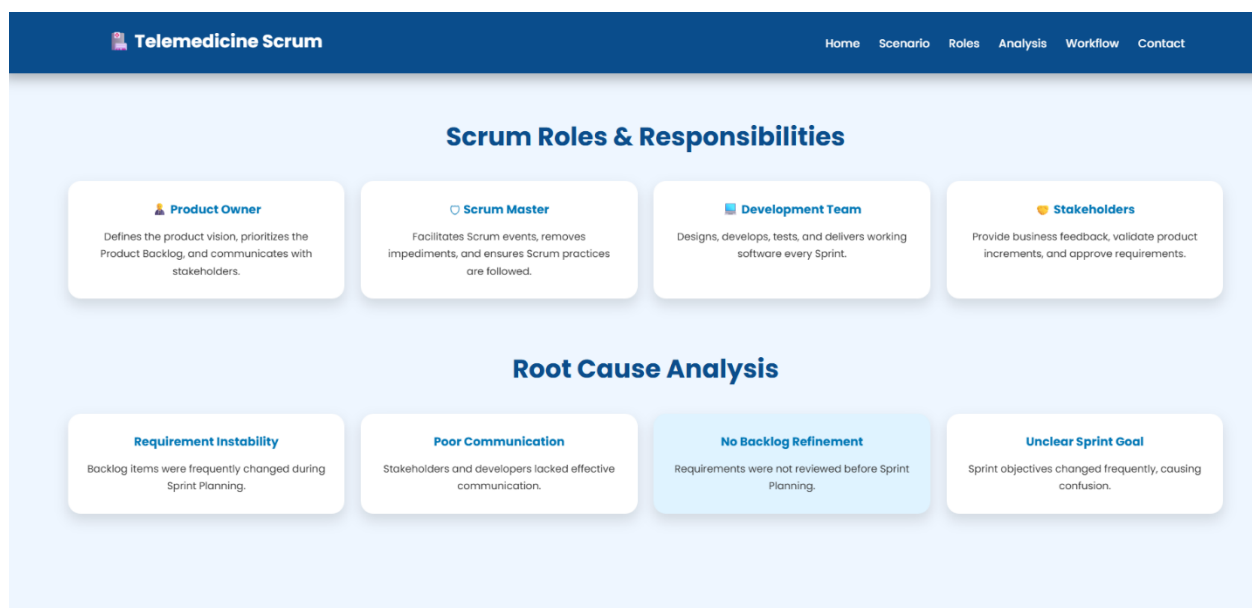
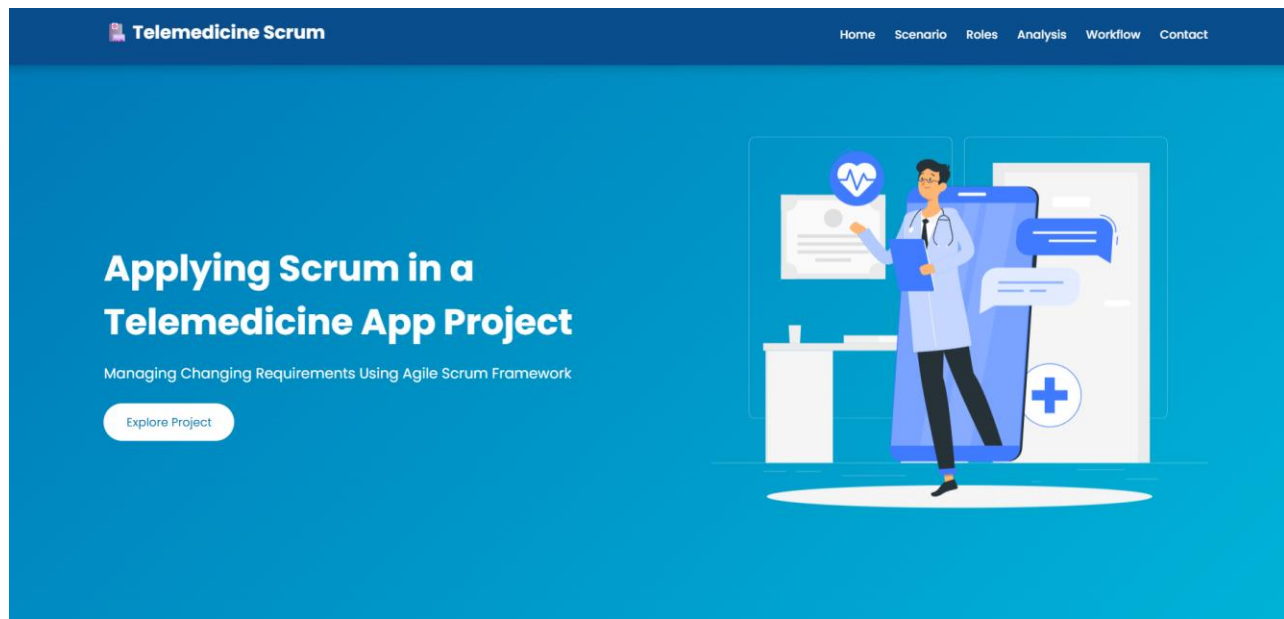
CSA1001-SOFTWARE ENGINEERING

ASSIGNMENT 1

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Problem 1: Applying Scrum in a Telemedicine App Project



Scrum Workflow

The Scrum process starts with Product Backlog creation, followed by Sprint Planning, Sprint Execution, Daily Scrum, Sprint Review, and Sprint Retrospective. This cycle continues until the product achieves customer satisfaction.



Improved Backlog Prioritization

Improved Backlog Prioritization



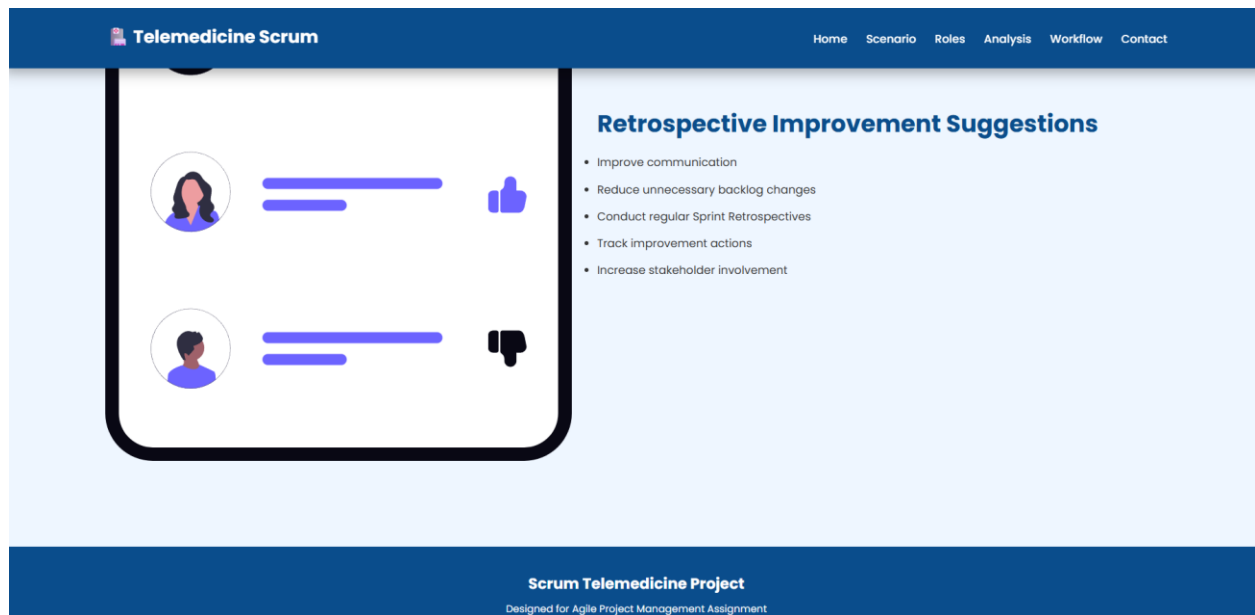
MoSCoW Prioritization

Must Have, Should Have, Could Have, Won't Have

Improvement Plan

- Weekly Backlog Refinement
- Freeze Sprint Backlog
- Story Point Estimation
- Stakeholder Review
- Clear Sprint Goal





Problem 2: Feasibility Study for a Smart Parking System

Smart Parking System

Feasibility Study for an IoT-Based Real-Time Parking Solution



Project Overview

The Smart Parking System uses IoT sensors and a mobile booking application to help drivers locate, reserve, and pay for parking spaces in real time. A feasibility study is conducted to determine whether the project is technically achievable, operationally practical, and financially beneficial.

Technical Feasibility

IoT Sensors

- Detect vacant parking slots
- Real-time occupancy updates
- Accurate vehicle detection

Mobile Application

- Parking reservation
- GPS navigation
- Digital payment integration

Cloud Database

- Store parking data
- User management
- Booking history tracking

Internet Connectivity

- Wi-Fi support
- 4G / 5G network
- Fast communication

Technical Feasibility Summary

The project is technically feasible because IoT sensors, cloud services, GPS technology, and mobile application frameworks are readily available. Existing APIs for maps and payments reduce development effort and improve scalability.

Operational Feasibility

Driver

- Find parking slots
- Reserve spaces
- Make online payments

Administrator

- Monitor occupancy
- Manage spaces
- Generate reports

Maintenance Team

- Monitor sensors
- Repair devices
- Maintain network

Operational Feasibility Report

The system is easy to operate for both drivers and administrators. Drivers can reserve parking spaces through a simple mobile app, while administrators manage parking activities through a centralized dashboard. Minimal training is required.

Financial Feasibility

Estimated Cost		Amount (USD)
IoT Sensors		\$5,000
Cloud Services		\$2,000
Mobile App Development		\$12,000
Installation		\$3,000
Maintenance		\$2,000
Total Cost		\$24,000

Benefit		Impact
Higher Parking Revenue		High
Reduced Traffic Congestion		High
Fuel Savings		Medium
Improved User Satisfaction		High

Design Thinking Activity

Empathy Interviews

Empathy interviews are conducted with drivers to understand their parking challenges, preferences, and expectations. This helps developers design a user-friendly and efficient parking application.

Questions Asked

- How long do you spend searching for parking?
- What difficulties do you face?
- Would you use online parking reservations?
- Which payment method do you prefer?
- What features would improve your experience?

The Smart Parking System is technically achievable, operationally practical, and financially beneficial. Conducting empathy interviews ensures that the final solution addresses real user needs and improves overall customer satisfaction.

Problem 3: Lean + XP in a Real-Time Tutoring Platform

Lean + XP in a Real-Time Tutoring Platform

Reducing Waste • Improving Quality • Faster Software Delivery



Project Overview

A software team developing a real-time tutoring platform aims to reduce waste using Lean Software Development and improve software quality through Extreme Programming (XP). Combining these agile approaches enables faster delivery, continuous improvement, and better customer satisfaction.

Lean Software Development – Seven Wastes

1. Waiting

- Delayed approvals
- Waiting for feedback
- Idle development time

2. Extra Features

- Unnecessary functions
- Unused modules
- Higher maintenance cost

3. Task Switching

- Frequent interruptions
- Reduced productivity
- Loss of focus

4. Rework

- Fixing repeated bugs
- Poor initial design
- Extra testing effort

5. Unused Knowledge

- Ignoring team ideas
- Poor communication
- Lost innovation

6. Extra Processes

- Too many meetings
- Excessive documentation
- Unnecessary approvals

7. Defects

- Software bugs
- Customer complaints
- Repeated fixes

Technical Summary

Lean Software Development focuses on eliminating activities that do not add value to the customer. By reducing delays, unnecessary work, defects, and inefficient processes, the tutoring platform can be delivered faster with improved quality and lower development cost.

Extreme Programming (XP) Practices

👤 Pair Programming

- Two developers work together.
- Better code quality.
- Knowledge sharing.

📝 Test Driven Development

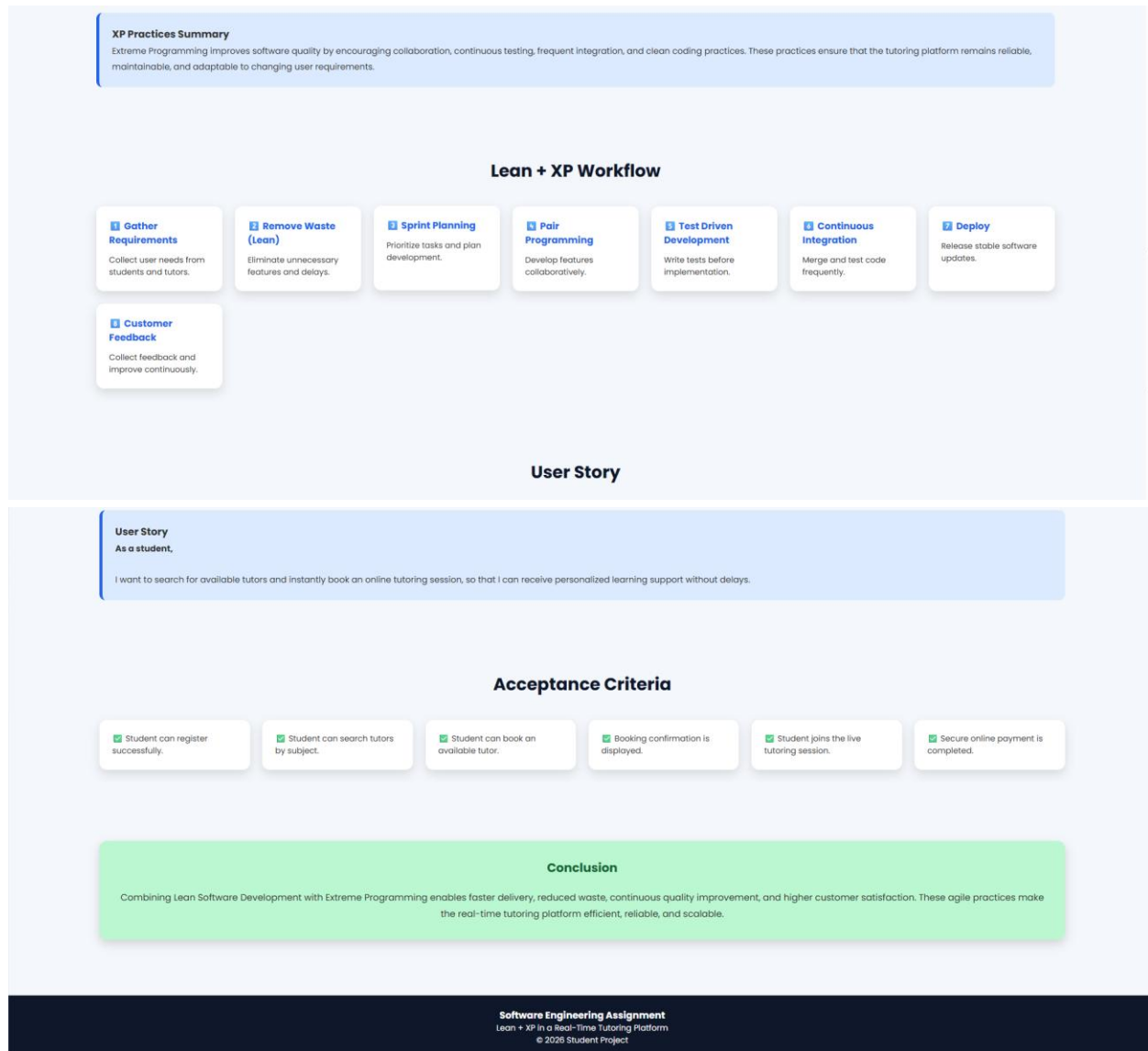
- Write tests before coding.
- Reduce software defects.
- Increase reliability.

🔄 Continuous Integration

- Merge code frequently.
- Automatic testing.
- Early bug detection.

🔧 Refactoring

- Improve code readability.
- Remove duplication.
- Maintain clean code.



Problem 4: Scaling Agile Using SAFe for Banking Software

Scaling Agile Using SAFe for Banking Software

Coordinating Multiple Agile Teams for Secure and Reliable Banking Solutions



Project Overview

A banking organization is developing a large-scale software system involving multiple teams such as mobile banking, payment processing, security, customer management, and database services.

The Scaled Agile Framework (SAFe) helps coordinate these teams, manage dependencies, align business goals, and deliver secure banking features incrementally.

SAFe Principles

Alignment

All Agile teams work toward common banking business goals and priorities.

Built-in Quality

Security, testing, compliance, and quality are included throughout development.

Transparency

Teams and management can track progress, risks, issues, and dependencies.

Program Execution

Multiple Agile teams coordinate their work through a common plan.

Continuous Improvement

Teams regularly inspect their work and improve their processes.

SAFe Roles in Banking

Product Owner

- Manages team backlog
- Defines user stories
- Prioritizes features

Scrum Master

- Facilitates Agile events
- Removes team obstacles
- Supports improvement

Product Manager

- Defines product vision
- Manages features
- Coordinates requirements

Release Train Engineer

- Coordinates multiple teams
- Manages dependencies
- Facilitates PI Planning

System Architect

- Defines architecture
- Ensures scalability
- Guides technical decisions

Banking Agile Teams

Mobile Banking Team

Develops mobile banking features such as account viewing, transfers, and notifications.

Payment Team

Handles payment processing and transaction management.

Security Team

Manages authentication, authorization, encryption, and security controls.

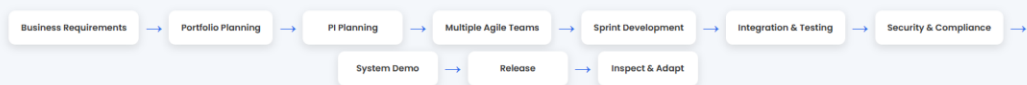
Customer Management Team

Handles customer profiles, accounts, and customer-related services.

Database Team

Manages banking data, database performance, and reliability.

SAFe Banking Workflow



What is PI Planning?

PI Planning is a collaborative planning event where multiple Agile teams discuss business objectives, identify dependencies, plan features, and agree on the work for the next Program Increment.

Banking Example: Instant Mobile Payments

Suppose a bank wants to introduce an instant mobile payment feature. Different teams coordinate their responsibilities to deliver the feature successfully.

Banking Team	Responsibility
Mobile Team	Develop payment interface
Payment Team	Process transactions
Security Team	Authentication and security
Database Team	Store transaction records
Compliance Team	Regulatory checks

Benefits of SAFe

Better Coordination

Multiple teams work toward the same business objective.

Faster Delivery

Banking features can be developed and delivered incrementally.

Improved Security

Security and compliance are considered throughout development.

Dependency Management

Teams identify and manage dependencies before they become major problems.

Business Alignment

Software development remains connected to banking business goals.

Risks and Mitigation

Risk	Mitigation
Team Dependencies	Use PI Planning and dependency tracking
Security Vulnerabilities	Continuous security testing
Changing Requirements	Prioritized and regularly reviewed backlog
Communication Gaps	Scrum of Scrums and regular coordination
Integration Problems	Continuous integration and automated testing

Conclusion

SAFe provides a structured approach for scaling Agile across multiple banking software teams. Through PI Planning, coordinated Agile teams, continuous integration, built-in quality, security practices, and continuous improvement, banks can deliver secure and reliable software while maintaining strong business alignment.

Problem 5: Ethical Evaluation of an AI Hiring System

Ethical Evaluation of an AI Hiring System

Fairness • Transparency • Privacy • Accountability



Project Overview

An organization wants to use an AI-based hiring system to screen resumes, rank candidates, and recommend applicants for interviews. Before deployment, the system must be evaluated for fairness, transparency, privacy, accountability, and potential discrimination.

Ethical Principles

Fairness

The AI should evaluate candidates using job-relevant qualifications rather than irrelevant or protected characteristics.

Transparency

Candidates and recruiters should understand how the AI system is used and what factors influence recommendations.

Privacy

Candidate resumes and personal information must be securely collected, processed, and stored.

Accountability

Human decision-makers should remain responsible for final hiring decisions.

Non-Discrimination

The system should be tested for unfair patterns that could disadvantage groups of candidates.

Ethical Risks

Bias

- Historical hiring data may contain bias.
- AI may reproduce existing patterns.
- Some candidates could be unfairly disadvantaged.

Lack of Explainability

- Candidates may not understand decisions.
- Recruiters may not understand rankings.
- Trust can be reduced.

Privacy Risks

- Resumes contain personal information.
- Unauthorized access may expose data.
- Data must be protected.

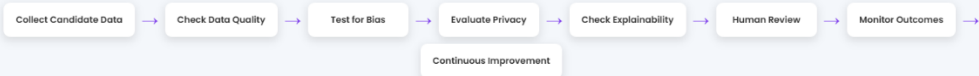
Automation Bias

- Recruiters may rely too heavily on AI.
- Human judgment may be reduced.
- Incorrect recommendations may be accepted.

Data Quality

- Incomplete training data.
- Unrepresentative datasets.
- Unreliable recommendations.

Ethical Evaluation Workflow



Bias Evaluation

Fairness Testing

The development team should evaluate whether the AI system produces substantially different outcomes for different groups. Testing should use appropriate privacy safeguards and legally permitted data.

Evaluation	Purpose
Selection Rate	Check whether recommendation rates differ substantially between groups.
False Positive Rate	Identify potentially unfair recommendations.
False Negative Rate	Check whether qualified candidates may be missed.
Model Performance	Verify that the system performs consistently.
Regular Audits	Detect fairness problems over time.

Bias and Ethical Risk Mitigation

Improve Training Data

Use high-quality and representative data while identifying problematic historical patterns.

Regular Bias Testing

Test the system before deployment and periodically after deployment.

Human Oversight

Recruiters should review AI recommendations rather than blindly accepting them.

Explainability

Provide understandable reasons for important recommendations where feasible.

Data Protection

Limit access to candidate data and retain information only as necessary.

Human-in-the-Loop Decision Process



AI should support recruiters rather than independently determine who gets hired. Human review provides an opportunity to identify inappropriate recommendations, missing information, or potential bias.

Ethical Decision Matrix

Ethical Issue	Risk Level	Recommended Action
Bias	High	Regular fairness audits
Privacy	High	Strong data protection
Lack of Transparency	Medium	Explainable decisions
Automation Bias	Medium	Human review
Poor Data Quality	High	Data validation

Recommendation: Controlled Deployment

The AI hiring system should not be used as a fully autonomous decision-maker.

It should be used as a decision-support tool only after appropriate bias testing, privacy assessment, security controls, explainability checks, human oversight, and continuous monitoring.

Conclusion

An ethical AI hiring system must be fair, transparent, privacy-preserving, and accountable. Human oversight, continuous auditing, and careful data management are essential to reduce discrimination and protect candidates.